Hanlong Liu

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EDUCATION

Georgia Institute of Technology — **Atlanta, GA**
Master of Science in Electrical and Computer Engineering(GPA: 4.0/4.0)
Master of Science in Computational Science & Engineering — *Expected Graduation:* 05/2027

The University of Michigan, Ann Arbor — **Ann Arbor, MI**
Bachelor of Science in Engineering (Computer Science Engineering) (GPA: 3.73/4.0) — *08/2022 - 05/2025*
Minor in Entrepreneurship (Ross School of Business)

PATENTS

- Jain, D., **Liu, H.** *Earphone and Method of Use.* US Patent App. 63/706, 317 filed Oct 2024
- – Developed a real-time DSP hardware pipeline for multi-driver transducer control, including filtering, impedance matching, adaptive gain control, and cross-driver synchronization.
- **Hanlong Liu**, *A Kind of armature driver Five Unit Hi-Fi Earphone*, ZL 2019 2 0331833.4
- – Designed system-level architecture for electromechanical driver arrays, optimizing latency, power usage, and signal integrity.

INTERNSHIP

ByteDance – Lark (Enterprise Collaboration Platform) — **Shenzhen, China**
Intern, AI Systems & Performance Engineering — *05/2023 – 08/2023*

- Optimized end-to-end AI system performance by profiling GPU kernels, memory bandwidth, and compute utilization across A100 and RTX 3090 platforms, achieving 10–25% higher hardware utilization through precision tuning and parallel scheduling strategies.
- Conducted multi-GPU training and inference optimization using tensor/model parallelism, reducing memory overhead and improving throughput under high-load environments.
- Tuned distributed inference pipelines by adjusting batching strategy, token allocation, and device-level parallel configurations, resulting in 15–30% latency reduction and 20–35% throughput improvement.
- Benchmarked custom inference stack against HuggingFace reference implementations, identifying bottlenecks in KV-cache access patterns, memory transactions, and GPU occupancy, delivering ~20% overall efficiency gain.
- Built and deployed an end-to-end LLM-powered inference service for a financial institution, supporting high-concurrency, low-latency online workloads through optimized batching, caching, and model serving strategies.

Development of a Patented Earphone Embedded Bone Conduction Technology — **The University of Michigan, Ann Arbor**
Research Intern, supervised by Prof. Dhruv Jain, Soundability Lab — *03/2024- Present*

- Developed a multi-unit earphone hardware architecture integrating bone-conduction actuators, balanced armature drivers, dynamic speakers, and electrostatic units, achieving high-fidelity reproduction across wide frequency ranges.
- Designed DSP signal-processing pipelines (filtering, calibration, cross-driver phase alignment) to improve audio stability across heterogeneous transducer types, reducing distortion and improving reliability in real-time scenarios.
- Built a 3-pass structured-light 3D reconstruction pipeline using FaceID depth data for anatomical ear modeling; performed point-cloud fusion, mesh alignment, and surface fitting to generate manufacturable ear-coupling geometries.
- Implemented CAD automation for custom ear-device shells, converting anatomical meshes into parametric manufacturable models through rule-based feature generation, reducing manual modeling effort by 70%+.
- Prototyped bone-conduction + dynamic hybrid Bluetooth earphones (Qualcomm QCC3040), integrating PCB layout, acoustic chamber design, and hardware–software co-development for real-world testing.

Voice to 3D — **Shenzhen, China**
Intern, Hardware Systems & 3D Modeling — *05/2023 - 09/2023*

- Incorporated Natural Language Processing into 3D modeling technologies to streamline usage and improve accessibility
- Employed a Transformer-based voice recognition model and integrated it with a NURBS-based 3D modeling system
- Prototyped a titanium alloy earphone design and led a strategic partnership with Tianqi Group

TECHNICAL SKILLS

- **Languages**: C, C++, Python, Verilog
- **Embedded & Systems**: real-time audio systems, hardware–software integration, DSP pipeline development
- **Circuit & Audio Systems**: Digital/analog circuit design, multi-driver transducer control, impedance matching, adaptive gain control, audio signal processing, latency & power optimization
- **PCB Design**: Schematic design, PCB layout, signal integrity optimization,
- **DSP & Analysis Tools**: Audio spectrum analysis, filtering & compensation design, MATLAB, DSP development/debug tools
- **CAD & 3D Design**: Rhino 3D, AutoCAD, , CAD automation for manufacturable geometry, 3D-printing workflows
- **Lab Equipment**: Oscilloscope, Spectrum Analyzer, signal generator, audio measurement rigs, impedance testing tools
- **Developer Tools**: Git, Linux/Unix systems, hardware debugging tools
- **Specialized**: Hardware–software co-design, acoustic system optimization, embedded DSP pipelines, manufacturability-oriented CAD